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$$\rho \text{ is PURE} \iff \begin{cases} \text{Tr}(\rho) = 1 \\ \text{Tr}(\rho^2) = 1 \end{cases}$$

$$\text{e.g. } \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \quad \rho^2 = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/4 \end{bmatrix}$$

$$\downarrow$$

$$\text{Tr}(\rho^2) = 1/4 + 1/4 = 1/2 \neq 1$$

$$\Rightarrow \text{MIXED STATE}$$

" $\Rightarrow$ "

$$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$\rho^2 = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$\text{Tr}(\rho^2) = 1/2 + 1/2 = 1$$

$$\text{PURE}$$

" $\Leftarrow$ "

$$\rho = \sum_K \lambda_K |K\rangle\langle K| \quad \forall \lambda_K \geq 0$$

$$\text{WE ASSUME } \cdot \text{Tr}(\rho) = \sum_K \lambda_K = 1 \quad \left| \cdot \left(\sum_K \lambda_K \right)^2 = \sum_K \sum_{K'} \lambda_K \lambda_{K'} \right.$$

$$\cdot \text{Tr}(\rho^2) = \sum_K \lambda_K^2 = 1 \quad \left| \cdot \sum_K \lambda_K^2 + \sum_{K \neq K'} \lambda_K \lambda_{K'} \Rightarrow \right.$$

$$\Rightarrow \sum_{K \neq K'} \lambda_K \lambda_{K'} = 0 \quad \text{AND ONLY 1 EIGENVALUE IS NOT 0}$$

EXAM QUESTION
ONE DOT PRODUCT
AND COMPUTE THE STATE

$$\text{STATE } \cdot |\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle - \frac{1}{2} |1\rangle \quad \rho_{\text{SUBST}} = \frac{1}{2} \begin{pmatrix} 1 + \omega_2 & \omega_2 - i\omega_1 \\ \omega_1 + i\omega_2 & 1 - \omega_2 \end{pmatrix}$$

TO KNOW

$$\text{COMPUTE } \rho = |\psi\rangle\langle\psi| = \left(\frac{\sqrt{3}}{2} |0\rangle - \frac{1}{2} |1\rangle \right) \left(\frac{\sqrt{3}}{2} \langle 0| - \frac{1}{2} \langle 1| \right) = \frac{3}{4} |0\rangle\langle 0| + \frac{1}{4} |1\rangle\langle 1| - \frac{\sqrt{3}}{4} |1\rangle\langle 0| - \frac{\sqrt{3}}{4} |0\rangle\langle 1|$$

$$\rho_{00} = \langle 0 | \rho | 0 \rangle = \frac{3}{4} = \frac{1}{2} (1 + \omega_2); \quad \omega_2 = 1/2$$

$$\rho_{01} = \langle 0 | \rho | 1 \rangle = -\frac{\sqrt{3}}{4} = \frac{1}{2} \omega_1; \quad \omega_1 = -\frac{\sqrt{3}}{2}$$

$$\omega \left(-\frac{\sqrt{3}}{2}, 0, \frac{1}{2} \right)$$

WHY $\omega_1 = 0$??

Consider a qubit prepared in the mixed state

$$p = \frac{1}{3} \quad |+\rangle$$

1. Find the **density matrix**

$$p = \frac{2}{3} \quad |0\rangle$$

2. Find the **Bloch vector**

DENSITY MATRIX

$\mathcal{B} \{ |0\rangle, |1\rangle \}$

$$\rho = \frac{1}{3} |+\rangle\langle +|$$

$$\rho = \frac{2}{3} |0\rangle\langle 0|$$

$$\rho = \frac{1}{3} |+\rangle\langle +| + \frac{2}{3} |0\rangle\langle 0|$$

$$\rho = \frac{1}{3} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{6} + \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} \end{bmatrix} = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} \end{bmatrix}$$

$$\frac{1}{6} (1 - \omega_z) = \frac{1}{6} \rightarrow \omega_z = \frac{2}{3}$$

$$\frac{1}{6} \omega_z = \frac{1}{6} \rightarrow \omega_x = \frac{1}{3}$$

$$\rho = \frac{1}{2} \begin{bmatrix} 1 + \omega_z & \omega_x - i\omega_y \\ \omega_x + i\omega_y & 1 - \omega_z \end{bmatrix}$$

$$\omega = (\frac{1}{3}, 0, \frac{2}{3})$$

$$\downarrow$$

$$|\omega|^2 = \frac{1}{9} + \frac{4}{9} = \frac{5}{9} < 1 \quad \checkmark$$

↑ must be less than 1

$$\rho(\epsilon) = |\psi(\epsilon) \times \psi(\epsilon)| \quad |\psi(\epsilon)\rangle = U(\epsilon) |\psi(0)\rangle$$

$$\langle \psi(\epsilon) | = \langle \psi(0) | U(\epsilon)^\dagger$$

$$\rho(\epsilon) = U(\epsilon) |\psi(0)\rangle \underbrace{\langle \psi(0) | U(\epsilon)^\dagger}_{=1} = U(\epsilon) \rho(0) U(\epsilon)^\dagger$$

$$\frac{d}{d\epsilon} \langle A \rangle = \frac{1}{i\hbar} \langle [A, H] \rangle + \left\langle \frac{dA}{d\epsilon} \right\rangle$$

$$A = \rho \quad \rho = |\psi \times \psi| \quad \text{PURE STATE}$$

$$\frac{d}{d\epsilon} \langle \rho \rangle = \frac{1}{i\hbar} \langle [\rho, H] \rangle + \left\langle \frac{d\rho}{d\epsilon} \right\rangle$$

$$\langle \rho \rangle = \langle \psi | \psi \times \psi | \psi \rangle = 1$$

$$\frac{1}{i\hbar} \langle [\rho, H] \rangle + \left\langle \frac{d\rho}{d\epsilon} \right\rangle \Rightarrow \left\langle \frac{1}{i\hbar} [\rho, H] + \frac{d\rho}{d\epsilon} \right\rangle = 0$$

IF YOU SWAP 12 THE USUAL ρ -

$$\Rightarrow \frac{1}{i\hbar} \overbrace{[\rho, H]}^{\text{swap}} + \frac{d\rho}{d\epsilon} = 0 \Rightarrow \frac{d\rho}{d\epsilon} = \frac{1}{i\hbar} [H, \rho]$$